

# Impact of vegetation restoration on preferential flow and soil infiltration capacity in the hilly region of the Loess Plateau

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## ABSTRACT

**Study region:** The Zhifanggou Watershed, a typical watershed in the hilly region of Loess Plateau in China.

**Study focus:** Assessing effects of various vegetation types on soil water infiltration facilitates understanding of soil hydrological processes and provides valuable references for reforestation in arid areas. Double-ring infiltrometers and dye tracing were employed across plots with various vegetation types and restoration durations to comprehensively investigate the impact of vegetation restoration on soil infiltration capacity, preferential flow, and their influencing factors.

**New hydrological insight for the region:** The results showed that the highest stable infiltration rate occurred in 30-year-old replanted monoculture forestland ( $0.57 \text{ cm min}^{-1}$ ), followed by 30-year-old replanted shrubland ( $0.42 \text{ cm min}^{-1}$ ). Monocultured forestland had slightly higher soil infiltration rates than mixed forestland, with macropore flow was dominant. Mixed forestlands exhibited the highest degree of preferential flow and the most preferential flow paths. Soil infiltration capacity and preferential flow paths increased with restoration years, and preferential flow was promoted in deeper soil layers. Soil macroporosity, clay content, and root length density predominantly influenced soil infiltration capacity and dye-stained characteristics parameters. Root characteristics were critical for the degree of preferential flow, with rock fragment and cracks also being not negligible factors. These findings could deepen our understanding of vegetation restoration in improving soil hydrological functions, and provide a reference for the management of vegetation restoration.

## 1. Introduction

Soil infiltration is an essential hydrological process that transforms atmospheric, surface, and groundwater, redistributes soil moisture, and reduces runoff, erosion, and evaporation (Bormann and Klaassen, 2008). It provides essential moisture for vegetation and crops, influences nutrient cycling and energy flow. Consequently, soil infiltration serves an essential function in water balance, soil

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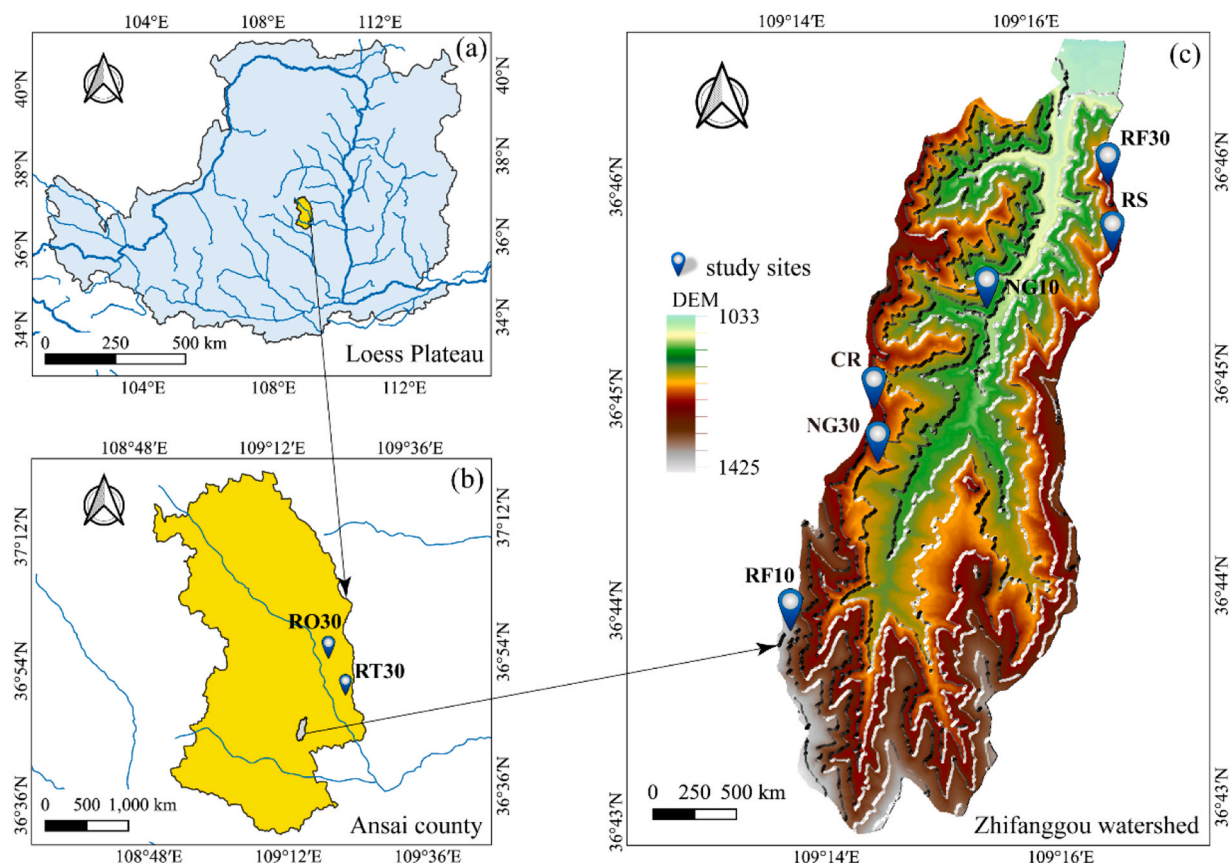
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health, and ecosystem dynamics, significantly affecting water quality, vegetation growth, and the occurrence and intensity of droughts and floods.

Soil infiltration replenishment primarily exists in the form of a matrix and preferential flows. Matrix flow describes the comparatively slow movement of water and solutes through the soil in a uniformly advancing wetting front that moves downward layer-by-layer. Conversely, the flow of water that bypassing the majority of the soil matrix and rapidly moving through large vertical or lateral pores is referred to as preferential flow. The contribution of preferential flow paths to total infiltration ranges from approximately 11–94 % (Alaoui, 2015), whereas the complex water movements associated with infiltration events are challenging to evaluate in field (Allaire et al., 2009). Recently, preferential flow has been studied using invasive techniques and by directly observing disturbed soil profiles (Abou Najm et al., 2010). Dyes are typically utilized to follow the pathways of water movement (Ghodrati and Jury, 1990), and Brilliant Blue FCF, owing to its low adsorption and high mobility, has been employed by various researchers as a dye tracer to visualize flow patterns in soil (Dusek and Vogel, 2018; Hou et al., 2023). Dyes can be applied to the soil surface (Gerke et al., 2015) or applied in the course of infiltration experiments using infiltrometers. Although dye patterns show only one cumulative flow path in experiment results, they are cost-effective and provide high spatial resolution (Flury et al., 1994; Yasuda et al., 2001). Images of dyed soil profiles can offer direct details regarding the depth and continuity of the dye using image analysis.

Alterations in soil infiltration are influenced by different vegetation types across various climatic zones. Lozano-Baez et al. (2019) demonstrated that afforestation in tropical and subtropical regions enhances soil infiltration. Huang et al. (2017) highlighted the advantages of leguminous grasslands in promoting precipitation infiltration into arid zones. Zhu et al. (2020) found that shrub communities exhibit better infiltration properties than grass communities in arid and semi-arid regions. Moreover, the time of restoration also affects infiltration. Leung et al. (2018) identified the critical role of plant age in soil infiltration rates during the early phase of plant development. Plant–soil interactions may change during the restoration process. The age and successional stage of plants affect these interactions, and longer rhizosphere interactions strengthen coupling (De Deyn et al., 2008).

The evidence suggests that vegetation restoration has a substantial impact on soil infiltration processes owing to alterations in soil and vegetation characteristics. The physiochemical properties of soil vary among various vegetation types (Wen et al., 2021). Changes in land use following vegetation restoration can modify plant root systems and nitrogen fixation, leading to shifts in soil microbial



**Fig. 1.** Overview of the study area, showing the location of Ansa County (a) and study sites (b)–(c). CR: cropland; NG10: natural grassland for 10 years; NG30: natural grassland for 30 years; RF10: replanted monoculture forestland for 10 years; RF30: replanted monoculture forestland for 30 years; RT30: replanted mixed forestland with *Robinia pseudoacacia* and *Pinus tabuliformis* for 30 years; RO30: replanted mixed forestland with *Robinia pseudoacacia* and *Platycladus orientalis* for 30 years; RS30: replanted shrubland for 30 years.

community richness and composition, which in turn alter the physical properties of the soil (Zhang et al., 2021). Changes in soil surface properties inevitably modify other soil properties. Studies have found that vegetation restoration notably alters the soil properties including soil bulk density, porosity, organic matter, moisture content, and texture (Gu et al., 2019; Zhang et al., 2022). The infiltration of soil is affected by its properties, which modify soil structure and pore traits. These properties are intimately associated with soil texture and undergo changes due to soil compaction and aggregate formation (Chandler et al., 2018). Most researches have indicated that soil infiltration declined as bulk density and clay content increased (Kabir et al., 2020; Yi et al., 2021). In addition, the large pores formed by plant roots are of great importance in the downward movement of soil moisture, especially in water-scarce arid and semi-arid regions (Guo et al., 2019). The morphological structure of roots from different vegetation types directly influences the spatial distribution of root channels, which are closely associated with soil moisture infiltration. For example, fine roots determine the infiltration potential in semi-arid areas and deep-rooted plants can substantially enhance subsoil water storage (Cui et al., 2019). The decomposition of forest root systems also contributes significantly to soil infiltration and moisture replenishment (Wu et al., 2021). These channels, formed by plant root systems, serve as common preferential paths for water and solute transport in the soil.

The Loess Plateau has experienced significant increase in vegetation cover due to the implementation of the Grain-for-Green Project since 1999 (Fan et al., 2024). Vegetation restoration serves to control soil erosion and to enhance the ecological environment, but can also dry out the soil layer during excessive restoration (Wang et al., 2024a, 2024b). Moreover, rainfall infiltration represents the primary means of replenishing soil moisture on the Loess Plateau, and the scarcity of moisture in arid and semi-arid regions is recognized as a major constraint on the vegetation survival and growth. The emergence of preferential flow paths marks the transition of soil from homogeneous to heterogeneous (Lin, 2010) and improves deep soil moisture by regulating infiltration and enhancing soil moisture effectiveness in arid regions. In addition, variations in vegetation properties and restoration duration inevitably result in dissimilarities in soil and root properties, which further affect soil infiltration. However, few studies have combined different vegetation restoration years and vegetation types to ascertain the impact of vegetation restoration on soil infiltration capacity and preferential flow. Furthermore, the spatial variation in the preferential flow paths among varying types of vegetation restoration necessitates a quantitative assessment.

The objectives of this study are to (1) conduct a comprehensive assessment of the impact of alterations in soil properties and roots characteristics on soil infiltration capacity and preferential flow due to vegetation restoration, and (2) to identify the principal factors causing changes in soil infiltration and preferential flow.

## 2. Material and methods

### 2.1. Study area

The study was conducted in the Zhifanggou Watershed, located in Ansai County on the northern Loess Plateau of China. The watershed covers 8.27 km<sup>2</sup> with elevations of 1100–1400 m above mean sea level (Fig. 1, Table 1). The mean annual precipitation amounts to 482 mm while the mean annual air temperature reaches 8.8°C. The region is a typical loess hilly and gully area covered with a thick layer of silt-loam loess, which makes it susceptible to soil erosion. Since the initiation of the Grain-for-Green project, the ecological condition of the area has gradually improved. The study area exhibits a diverse range of vegetation types, with common populations of tree species, including *Robinia pseudoacacia* and *Populus simonii*; shrub species, including *Sophora vicifolia* and *Caragana korshinskii*; and grass species, including *Artemisia sacrorum*, *Artemisia giraldii*, and *Stipa bungeana*.

### 2.2. Experimental treatment

#### 2.2.1. Soil infiltration and dye tracer experiment

The soil infiltration experiment was conducted in the selected study plots during the summer of 2023 using a double-ring infil-

**Table 1**  
Detailed information of study sites.

| Land use type | Restoration period (years) | ID   | Vegetation type                  | Slope | Coverage (%) | Elevation (m) | Dominant species                                                 |
|---------------|----------------------------|------|----------------------------------|-------|--------------|---------------|------------------------------------------------------------------|
| Cropland      | -                          | CR   | Cropland                         | 0     | -            | 1249          | Corn                                                             |
| Grassland     | 10                         | NG10 | Secondary natural grassland      | 15    | 21.7         | 1195.07       | <i>Herba artemisiae</i>                                          |
|               | 30                         | NG30 | Secondary natural grassland      | 22    | 40.2         | 1227.1        | <i>Artemisia giraldii</i> Pamp;<br><i>Bothriochloa ischaemum</i> |
| Forestland    | 10                         | RF10 | Replanted monoculture forestland | 33    | 61.0         | 1321.32       | <i>Robinia pseudoacacia</i>                                      |
|               | 30                         | RF30 | Replanted monoculture forestland | 18    | 75.8         | 1305.83       | <i>Robinia pseudoacacia</i> ;<br><i>Caragana intermedia</i> .    |
|               | 30                         | RO30 | Replanted mixed forestland       | 35    | 75.0         | 1107.16       | <i>R. pseudoacacia</i> ;<br><i>Platycladus orientalis</i>        |
|               | 30                         | RT30 | Replanted mixed forestland       | 27    | 67.9         | 1282.46       | <i>Robinia pseudoacacia</i> ;<br><i>Pinus tabuliformis</i>       |
| Shrubland     | 30                         | RS30 | Replanted shrubland              | 19    | 43.5         | 1248          | <i>Caragana korshinskii</i>                                      |

tration instrument with two rings and a Mariotte bottle. Two rings with diameters of 15 and 30 cm were gently inserted into the soil at a depth of 20 cm after removing the litter from the soil surface to ensure that the sidewalls and edges were watertight. A 4 g L<sup>-1</sup> concentration solution of Brilliant Blue FCF was simultaneously added to the double rings at the same time to visualize the preferential flow paths. Mariotte bottles were used to keep the water head in the two rings constant at 8 cm. The water level in the Mariotte bottle was monitored at specific time intervals throughout the experiment until the infiltration rate stabilized. Each experiment lasted 60 min, with two replicates in each plot, resulting in a total of 18 experiments. The average values for the first 3 minutes and entire 60 minutes were the initial infiltration rate (IIR, the abbreviations are shown in Table 2), and average infiltration rate (AIR), respectively. The stable infiltration rate (SIR) is calculated as the average of the final three measured infiltration rates (Guo et al., 2019):

$$i = \Delta I / \Delta t \quad (1)$$

where  $i$  represents the infiltration rate (cm min<sup>-1</sup>) and  $\Delta I$  denotes the cumulative infiltration within the inner ring (cm) over the course of the infiltration time  $\Delta t$  (min).

The Mariotte bottles were closed upon completion of the 60-min experiment. The double rings were meticulously removed after the dye had fully permeated the soil. A vertical soil profile was excavated along the outer ring diameter. The vertical dye photographs were taken using the camera stationary and a calibrated steel strip was placed to position it along the horizontal and vertical axes of the stained region for reference.

### 2.2.2. Image and data analysis of preferential flow

The dye patterns were extracted from the soil profile images using Brilliant Blue infiltration experiments in Python 3.8, with scikit-image (skimage) and OpenCV libraries. This involved cropping and geometrically correcting of the raw image, binarizing it into pixels that were either stained or unstained, and then labelling of the extracted dye patches. In addition, only the dyed portion within the width of the inner ring was analyzed to focus on the vertical infiltration process. In view of the measured dye-staining depths in each excavated soil profile, dye-stained images with a depth of 45 cm and width of 15 cm were used to calculate the dye-stained characteristic parameters of each soil profile. This value was obtained by calculating the number of stained pixels. Due to the extensive collection of stained profiles, only one typical stained profile was selected from each plot for the subsequent analysis in this study. The dye-stained characteristic parameters were calculated from the dye images and include dye coverage (DC, %), defined as percentage of stained area coverage. Based on the average staining area percentage across the soil profile, the relative deviation (RD, %) of the staining area percentage at each depth was calculated to assess the spatial variability of staining areas at various depths. The maximum stained depth (MSD, cm) represents the bottom of water infiltration through the preferential flow paths.

The total dye coverage (TDC, %) was the ratio of the dye-stained area (TDA, cm<sup>2</sup>) to the entire profile. The depth at which the dye coverage was reduced to below 80 % was defined as the uniform infiltration depth (UniFr, cm), representing the depth at which matrix flow is prevalent (Van Schaik, 2009; Bargaes Tobella et al., 2014). The preferential flow fraction (PFFR, %) and length index (LI) described the degree and heterogeneity of preferential flow within the soil profile based on the above indices. The preferential flow fraction (PFFR, %) was described as the percentage of the total infiltration that followed the preferential flow path. The PFFR was calculated as follows:

$$PFFR = \left( 1 - \frac{UniFr \times W}{D_{Tot}} \right) \times 100\% \quad (2)$$

Here,  $W$  is the width of the soil vertical profile (15 cm) and  $D_{Tot}$  is the total dye-stained area (cm<sup>2</sup>). Accordingly, the length index (LI) of preferential flow was calculated as follows:

**Table 2**  
List of abbreviations in this paper.

| Abbreviations               | Full name                       | Abbreviations                                | Full name                     |
|-----------------------------|---------------------------------|----------------------------------------------|-------------------------------|
| <b>Soil properties</b>      |                                 | <b>Soil infiltration capacity</b>            |                               |
| GMD                         | Geometric mean diameter         | AIR                                          | Average infiltration rate     |
| IP                          | Soil ineffective porosity       | IIR                                          | Initial infiltration rate     |
| MAP                         | Soil macroporosity              | SIR                                          | Stable infiltration rate      |
| MWD                         | Mean weight diameter            | <b>Dye-stained characteristic parameters</b> |                               |
| MIP                         | Soil microporosity              | CV                                           | Coefficient of variation      |
| TP                          | Total porosity                  | LI                                           | Length index                  |
| SMC                         | Soil saturated water content    | MSD                                          | Maximum stained depth         |
| BD                          | Bulk density                    | PFFR                                         | Preferential flow fraction    |
| SWCC                        | Soil water characteristic curve | TDA                                          | Ratio of the dye-stained area |
| <b>Root characteristics</b> |                                 | TDC                                          | Total dye coverage            |
| RLD                         | Root length density             | UniFr                                        | Uniform infiltration depth    |
| RSA                         | Root surface area               |                                              |                               |
| <b>Statistical analysis</b> |                                 |                                              |                               |
| ANOVA                       | Analysis of variance            |                                              |                               |
| LSD                         | Least significant difference    |                                              |                               |

$$LI = \sum_n^{i=1} |DC_{i+1} - DC_i| \quad (3)$$

where  $DC_i$  is the dye-stained area of  $i$ , and  $i$  describes the soil depth (i.e., 1, 2, 3..., 45 cm). The coefficient of variation ( $CV_d$ ) refers to the coefficient of variation in the soil dyeing profile in this study, was calculated as follows:

$$CV_d = \frac{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (DC_i - \overline{DC})^2}}{\frac{1}{n} \sum_{i=1}^n DC_i} \quad (4)$$

where  $\overline{DC}$  represents the mean coverage of dye stains (Zhang et al., 2022).

To further determine the spatial distribution and number of preferential flow paths, we divided the dyed areas into multiple independent patches based on their spatial topological relationships in the dyed areas (Weiler and Flühler, 2004; Li et al., 2020). Each dyed area was defined as the number of pixels occupied by the plaque, approximated as rectangle, and the width was calculated based on the stained area. The stained patches of the vertically dyed image were divided into four size classes based on their widths: < 5.0, 5.0–10.0, 10.0–20.0, and > 20.0 mm. The coordinates of the center points of the stained patches were obtained.

### 2.3. Soil sampling and testing

All soil samples were taken from two randomly selected soil cores within each sampling plot at 20 cm intervals down to a depth of 40 cm, in an area adjacent to the dye site, before the application of the dye tracer. Undisturbed soil samples were collected using a cutting cylinder (volume: 100 cm<sup>3</sup>) and saturated by immersion in water until they reached full saturation, after which soil water characteristic curves (SWCCs) were measured using a high-speed centrifuge (CR21G). The Van Genuchten model was employed to fit the SWCCs. The soil saturated water content (SMC, %), ineffective porosity (IP, %), microporosity (MIP, %), and macroporosity (MAP, %) of the soil were determined from the SWCCs. Finally, soil samples were oven-dried for 24 h at 105°C until a constant weight was achieved, and the soil was weighed to calculate soil bulk density (BD, g cm<sup>-3</sup>) and total porosity (TP, %). Undisturbed soil samples were collected in tin containers and the wet sieving method was used to obtain the soil aggregate size distribution. Furthermore, calculations were performed for the mean weight diameter (MWD, mm) and geometric mean diameter (GMD, mm).

Disturbed soil samples were collected to measure soil particle size composition, organic matter content, and pH. Soil particle size distribution was assessed using a laser diffractometer (Mastersizer 2000, Malvern Instruments, Malvern, UK), with the contents of clay, silt, and sand differentiated according to the USDA classification. Soil organic matter content was assessed through the potassium dichromate colorimetric method. The soil pH was assessed with a glass electrode in suspension.

### 2.4. Root sampling and analysis

After each infiltration measurement, a specialized root-sampling tool (5 × 10 × 10 cm) was used twice to sample soil containing roots from depths of 0–40 cm at 20 cm intervals down the center of the vertical profile. These subsamples were rinsed with tap water through a 0.25 mm sieve for the laboratory assessment of root parameters. Fine (≤ 2 mm) and coarse roots (> 2 mm) were separated using Vernier calipers. Morphological parameters (length, area, and volume) of fine roots were measured using WinRHIZO (WinRHIZO Pro 2009b). The length and diameter of the coarse roots were measured using Vernier calipers (precision, 0.1 mm). The root surface area (RSA, cm<sup>2</sup>/500 cm<sup>3</sup>) was estimated and the total root length density (RLD, mm/cm<sup>3</sup>) was calculated as follows:

$$RLD = \frac{RL}{V} \quad (5)$$

where  $RL$  is the sum of the lengths of fine and coarse roots (mm) and  $V$  is the volume of the soil occupied by roots (cm<sup>3</sup>).

### 2.5. Statistical analysis

One-way analysis of variance (ANOVA) and the least significant difference (LSD) post hoc test were used to analyze the differences in soil physicochemical properties, root characteristics, and soil infiltration rates among the different vegetation types. Pearson's correlation coefficient was applied to represent the relationship between the dye-stained characteristic parameters, soil physicochemical properties, and root characteristics. The analyses were performed using R version 4.4.1.

## 3. Results

### 3.1. Changes in soil physicochemical properties and root characteristics

#### 3.1.1. Soil physicochemical properties

Many soil properties at 0–40 cm depth differed significantly among different vegetation types (Table 3). ANOVA revealed that



NG10 and RT30 had significantly higher BD than the others ( $P < 0.01$ ), with the highest BD at the 0–20 cm soil depth for NG10 ( $1.44 \text{ g cm}^{-3}$ ) and at the 20–40 cm for RT30 ( $1.45 \text{ g cm}^{-3}$ ). In contrast, the maximum TP were detected for RO30 (55.57 %) at 0–20 cm and for NG30 (58.38 %) at 20–40 cm depth. Except for NG10 and RT30, vegetation restoration increased TP ( $P < 0.01$ ) at 0–40 cm depth. The clay content of grassland (NG10 and NG30), CR, and RT30 was significantly higher than that of the others. The clay content peaked in the 0–20 cm layer of NG10 (17.60 %), whereas the highest silt content was observed at 20–40 cm depth for RT30 (66.91 %). The clay content of all tested plots declined at the 20–40 cm soil depth. The sand content of RF10, RS30, and RO30 were higher than that of CR ( $P < 0.05$ ). The highest value of SOM was observed for NG30, followed by CR. The pH of NG10, RF10, and RO30 significantly increased compared with that of CR. Additionally, NG30 exhibited the highest soil water stability among all vegetation types, significantly exceeding that of RF10, RF30, RS30, and RO30 at 20–40 cm soil depth.

As shown in Fig. 2, there were differences in water content among vegetation types near 0 bar, but these differences diminished as the pressure head increased. ANOVA demonstrated statistically significant differences ( $P < 0.05$ ) in macroporosity and ineffective porosity among different plots. The maximum macroporosity and ineffective porosity values were observed in RF30 (34.96 %) and RT30 (20.43 %), respectively. Compared with the CR, all vegetation types showed increased macroporosity and decreased ineffective porosity, except for RT30. The macroporosity of RF30, RO30, and RS30 was notably greater than that of CR. Vegetation restoration also improved microporosity except for NG10. RF30 had the highest microporosity value (7.84 %) followed by RS30 (7.78 %).

### 3.1.2. Root characteristics

The variations in root characteristics among distinct vegetation types at varying soil depths are shown in Fig. 3. At the 0–20 cm soil depth, the highest value of root length density was found in RF30 ( $223.38 \text{ mm cm}^{-3}$ ), followed by RO30 ( $183.38 \text{ mm cm}^{-3}$ ). The root length density of RF30 was markedly higher than those of NG10, RS30, and CR ( $P < 0.05$ ), and the root surface area of RO30 ( $162.27 \text{ cm}^2 / 500 \text{ cm}^3$ ) was significantly higher than those of NG10, RS30, RF10, and CR. The root length density ( $109.59 \text{ mm cm}^{-3}$ ) and root surface area ( $71.60 \text{ cm}^2 / 500 \text{ cm}^3$ ) of RF30 were greater than those of other vegetation types at 20–40 cm. In addition, the root systems of NG10 and RT30 were concentrated at 0–20 cm depth, while the root characteristics within CR were relatively poor across all soil depths.

### 3.2. Changes in soil infiltration after vegetation restoration

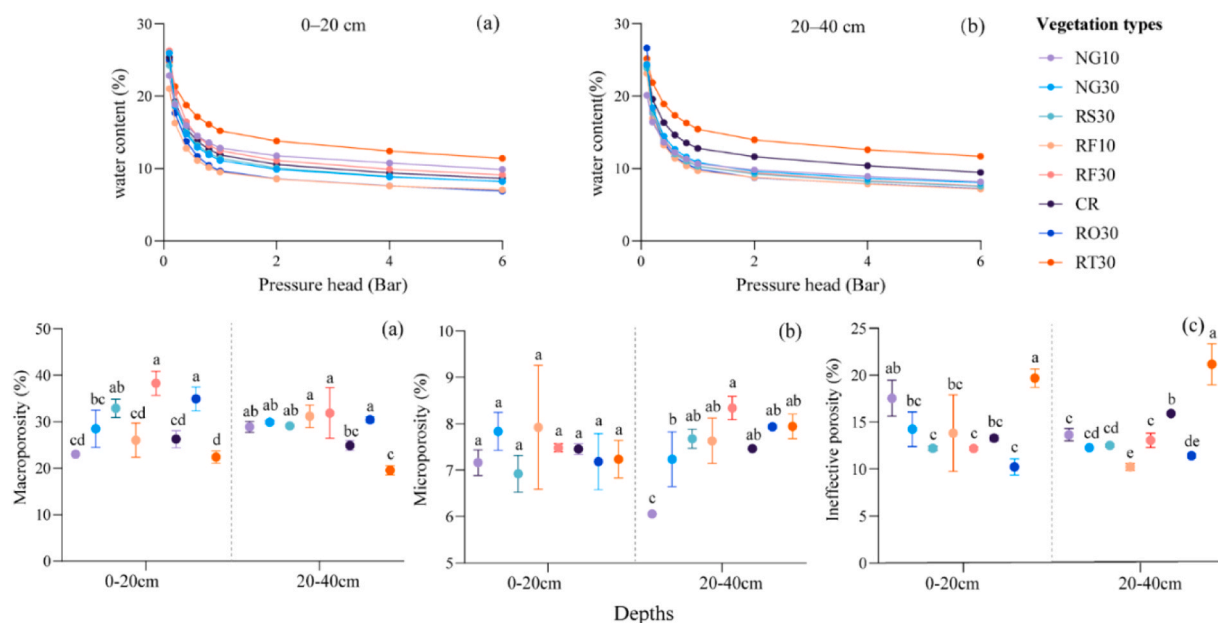
Vegetation restoration enhanced soil infiltration rates to varying extents (Fig. 4). In general, the infiltration rates of all plots showed a significant decline in the initial stage and then gradually leveled off. IIR, SIR, and AIR all increased in the soils of the various vegetation types, with these increases reaching significant levels in RF30 ( $P < 0.05$ ). Compared with CR, the IIR and SIR of mono-culture forestlands, shrublands, grasslands, and mixed forestlands were 192.31 %, 115.93 %, 66.48 %, and 109.18 % higher, and 333.33 %, 368.52 %, 175 %, and 103.70 % higher, respectively. The SIR of RF30 was the highest ( $0.57 \text{ cm min}^{-1}$ ), and was markedly ( $P < 0.05$ ) higher than that of CR ( $0.09 \text{ cm min}^{-1}$ ) and RT30 ( $0.13 \text{ cm min}^{-1}$ ). Additionally, the IIR of RF30 ( $2.55 \text{ cm min}^{-1}$ ) was notably greater in other types, except for RO30 ( $1.54 \text{ cm min}^{-1}$ ). The IIR of RF30 was approximately three times higher than that of the CR. The minimum value of AIR was recorded in CR at  $0.34 \text{ cm min}^{-1}$ , followed by NG10 at  $0.44 \text{ cm min}^{-1}$ , RT30 at  $0.59 \text{ cm min}^{-1}$ , and RF10 at  $0.66 \text{ cm min}^{-1}$ . These values were significantly lower than those observed for RO30, RS30, NG30, and RF30.

**Table 3**

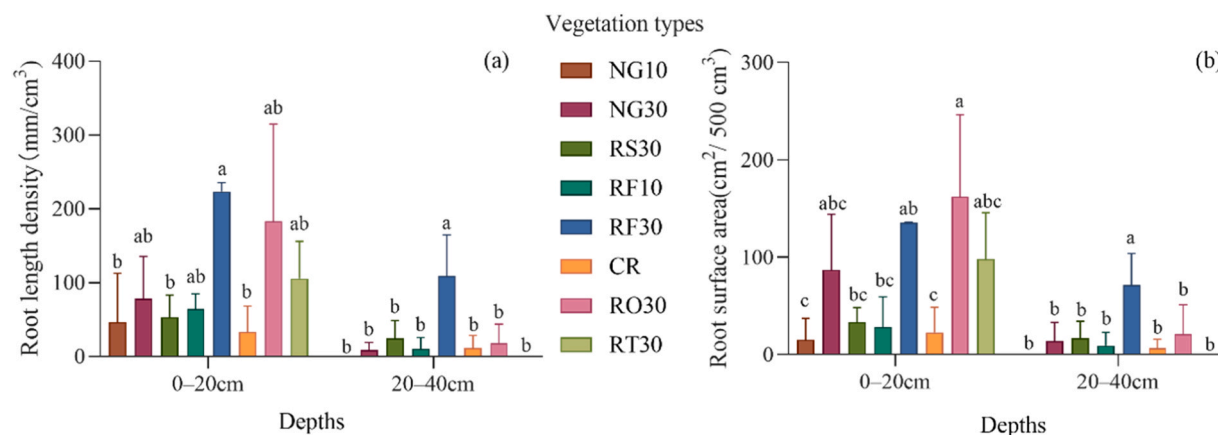
Mean values of soil physiochemical properties at 0–40 cm depth in various vegetation types.

| Vegetation types | Depth (cm) | BD ( $\text{g cm}^{-3}$ ) | TP (%)        | Clay (%)      | Silt (%)      | Sand (%)      | SOM ( $\text{g kg}^{-1}$ ) | pH           | MWD (mm)     | GMD (mm)      |
|------------------|------------|---------------------------|---------------|---------------|---------------|---------------|----------------------------|--------------|--------------|---------------|
| NG10             | 0–20       | <u>1.44a</u>              | 47.51c        | <u>18.96a</u> | 64.62ab       | 16.43d        | 11.34                      | <u>8.56a</u> | 1.21         | 0.49          |
|                  | 20–40      | 1.35b                     | 50.98b        | 16.24abc      | 61.31d        | 22.45ab       | 10.24ab                    | <u>8.63a</u> | 0.51ab       | 0.23b         |
| NG30             | 0–20       | 1.24bc                    | 54.92ab       | 14.82bcd      | 62.37abc      | 22.82bc       | <u>20.69</u>               | 8.31c        | <u>1.63</u>  | <u>0.63</u>   |
|                  | 20–40      | 1.22c                     | <u>55.57a</u> | 16.36abc      | 63.15bcd      | 20.50bc       | <u>13.73a</u>              | 8.42c        | <u>0.55a</u> | <u>0.23ab</u> |
| RS30             | 0–20       | 1.16c                     | 57.87a        | 12.94cde      | 61.04bc       | 26.02ab       | 16.88                      | 8.40abc      | 1.47         | 0.59          |
|                  | 20–40      | 1.31bc                    | 52.22ab       | 12.24bcd      | 63.21bcd      | 24.56ab       | 8.27ab                     | 8.43c        | 0.24c        | 0.15 cd       |
| RF10             | 0–20       | 1.20c                     | 56.47a        | 12.30de       | 60.03c        | <u>27.68a</u> | 10.94                      | 8.47ab       | 0.63         | 0.23          |
|                  | 20–40      | 1.32b                     | 51.89b        | 10.54d        | 61.90 cd      | <u>27.56a</u> | 5.24b                      | 8.61ab       | 0.22c        | 0.14d         |
| RF30             | 0–20       | 1.21c                     | 56.15a        | 11.13e        | 65.01ab       | 23.86bc       | 13.22                      | 8.35bc       | 1.24         | 0.40          |
|                  | 20–40      | 1.30bc                    | 52.65ab       | 11.98 cd      | 67.00a        | 21.01bc       | 8.30ab                     | 8.43c        | 0.37bc       | 0.18c         |
| CR               | 0–20       | 1.22c                     | 55.77a        | 15.75bc       | 62.23bc       | 22.02c        | 16.69                      | 8.44ab       | 0.58         | 0.24          |
|                  | 20–40      | 1.33b                     | 51.77b        | <u>18.17a</u> | 66.23ab       | 15.60c        | 10.71ab                    | 8.49bc       | 0.50ab       | 0.26ab        |
| RO30             | 0–20       | 1.14c                     | <u>58.38a</u> | 11.35e        | 62.98abc      | 25.67ab       | 13.52                      | 8.49a        | 1.32         | 0.42          |
|                  | 20–40      | 1.27bc                    | 53.92ab       | 11.46 cd      | 64.86abc      | 23.68ab       | 7.65ab                     | 8.60ab       | 0.31c        | 0.17 cd       |
| RT30             | 0–20       | 1.36ab                    | 50.43bc       | 17.70ab       | <u>66.66a</u> | 15.64d        | 15.18                      | 8.40abc      | 0.92         | 0.41          |
|                  | 20–40      | <u>1.45a</u>              | 47.14c        | 17.03ab       | <u>67.16a</u> | 15.81c        | 6.39ab                     | 8.44c        | 0.53a        | 0.26a         |

Note: Underlined entries indicate the highest values between the two depths. The presence of distinct lowercase letters above the bars denotes that there are significant differences across the various vegetation types within the same soil depth ( $P < 0.05$ )



**Fig. 2.** Soil water characteristic curves (a)–(b), macroporosity (c) microporosity(d), and ineffective porosity(e) of different vegetation types under various soil depths. The presence of distinct lowercase letters above the bars denotes that there are significant differences across the various vegetation types within the same soil depth ( $P < 0.05$ ).



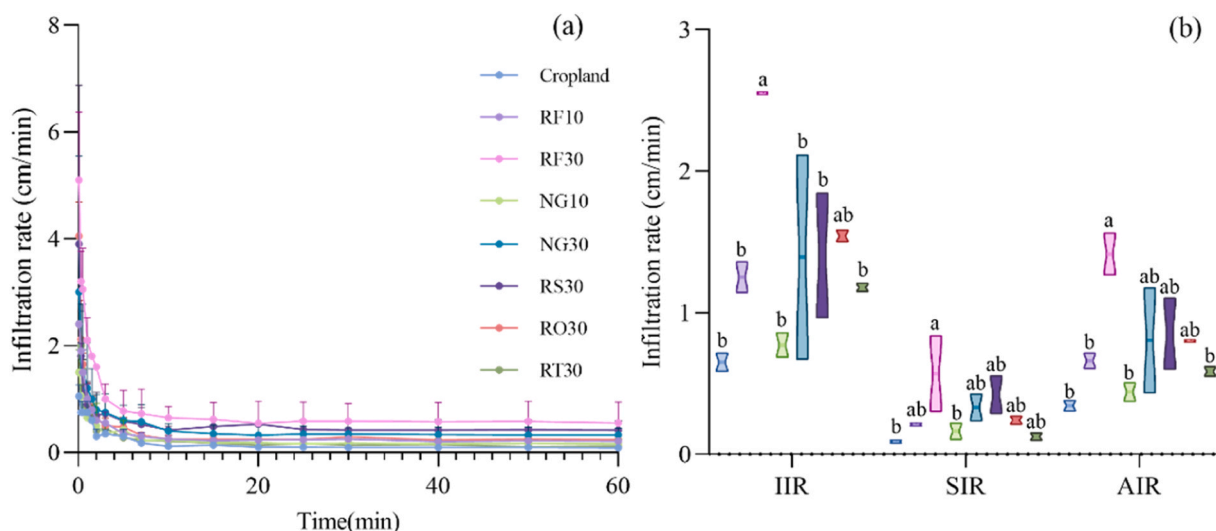
**Fig. 3.** Root characteristics across distinct vegetation types at various soil depths. (a) Root length density; (b) root surface area. The presence of distinct lowercase letters above the bars denotes that there are significant differences across the various vegetation types within the same soil depth ( $P < 0.05$ ).

### 3.3. Differences in dye-stained characteristics after vegetation restoration

#### 3.3.1. Dye-stained morphological traits

As shown in Fig. 5, the dye-stained characteristics parameters of different plots indicated that vegetation restoration promoted more extensive vertical movement. The water flow paths observed in various plots revealed that the lower boundary line were generally smoother in most plots, while those in the mixed forestlands displayed multiple peaks. Notably, longer 'dendritic' preferential flow was observed in RO30. Additionally, the vertical dye photographs showed that plant roots and macropores increased after vegetation restoration. Cracks and rock fragments were found distributed in the cropland and RT30.

Overall, the deeper soil layers exhibited a reduced DC (Table 4). The dye was distributed uniformly at the soil depth of 0–10 cm, whereas below 10 cm, the dye coverage sharply reduced progressively with deeper soil layers. This trend is further confirmed by the variation in the RD value, which also reflects a significant spatial difference in the dye-stained area across the soil profile. Compared with the CR, the total ratio of the dye-stained area (TDA) for monoculture forestlands, shrubland, grasslands, and mixed forestlands increased by 150.19 %, 102.15 %, 76.17 %, and 71.09 %, respectively. The maximum stained depth (MSD) was 42.08 cm for RF30,



**Fig. 4.** Changes in infiltration rates. (a) Changes in the infiltration rate over time; (b) Variations in IIR, SIR, and AIR across different vegetation types. The presence of distinct lowercase letters above the bars denotes that there are significant differences across the various vegetation types within the same soil depth ( $P < 0.05$ ).

35.80 cm for RF10, and only 16.83 cm for CR, confirming the highest soil infiltration rates in RF30 among all vegetation types. Similarly, the highest total dye-stained coverage (TDC) value was observed for RF30 (88.20 %), followed by RF10 (76.04 %), NG30 (66.78 %), and RS30 (66.35 %). Vegetation restoration enhanced connectivity and continuity of soil water movement. The uniform infiltration depth (UniFr) of RF30 was the deepest among all vegetation types, reaching approximately 37.50 cm, which suggested a stronger soil-water interaction. However, compared with CR, the average coefficient of variation (CV) for RF30 decreased by 76.37 %. The highest CV was observed for CR (135.86 %), which can be linked to the presence of cracks within the vertical profile of the cropland. Mixed forestlands exhibited the highest preferential flow fraction and length index values, with RO30 at 22.18 % and 412.94, respectively, followed by RT30 at 16.82 % and 396.47.

### 3.3.2. Spatial distribution of preferential flow paths

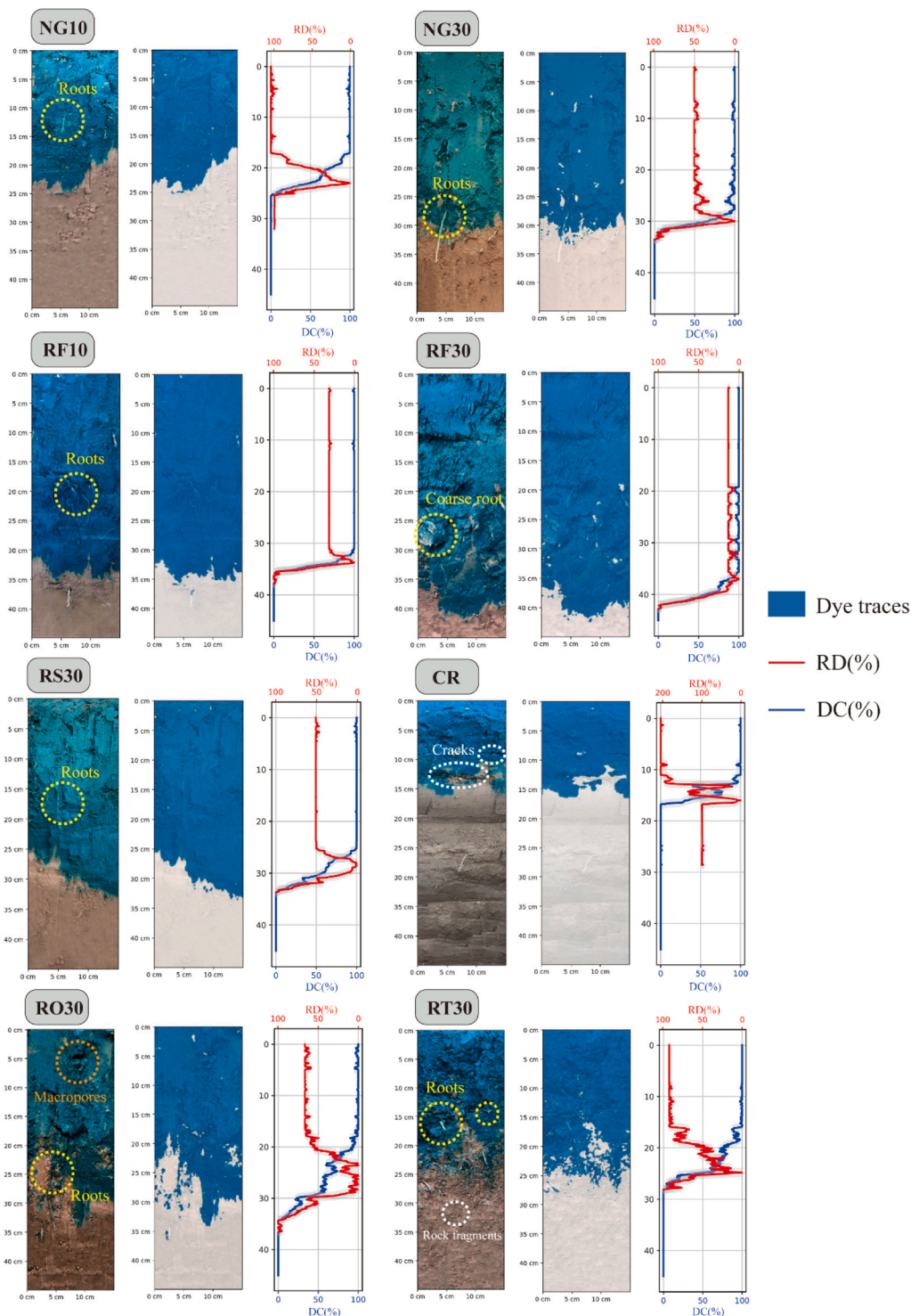
Fig. 6 illustrates wider stained patches ( $W > 20.0$  mm) in the CR were the shallowest (around 7.47 cm). Compared with NG10, CR, and RS30, the centers of the wider stained patches in the replanted monoculture forestlands and NG30 occurred deeper in the soil layers. Notably, wider stained patches were only observed in the surface soil, where  $> 80$  % of the area was stained, making it impossible to precisely obtain information regarding the preferential flow path in that layer. Therefore, the preferential flow paths were further distinguished below the surface soil, combined with the width and central position coordinates of the stained patches. Overall, the distribution of preferential flow paths with a width of  $< 5.0$  mm were the most extensive and densest, while those of 5.0–20.0 mm was relatively dispersed across all soil profile. The preferential flow paths in NG10 were primarily concentrated at 20–25 cm depth, whereas the highest aggregation was observed in the deeper layer (30–35 cm) in NG30. A similar pattern was observed for RF10 and RF30, suggesting that the preferential flow paths tended to aggregate in deeper layers with increasing restoration years. Additionally, a dense horizontal distribution was evident in the 30–35 cm soil layer of NG30. In RO30, the preferential flow paths were more concentrated in the vertical direction and exhibited diversity in width, leading to the formation of pronounced areas of larger aggregation. In contrast, the  $< 5.0$  mm paths were more widely distributed but dispersed in RT30.

Overall, the average number of stained patches in different dye-stained profiles exhibited that mixed forestland exhibited the greatest number of preferential flow paths, while CR had the lowest (Fig. 7). The wider stained patches were mostly concentrated within the 0–15 cm soil depth, with the largest proportion in the CR (25 %). The majority of the preferential flow paths ( $W < 5$  mm) were found below 15 cm depth. In RO30, the number of preferential flow paths with a width of  $5 < W < 10$  mm was the highest, totaling seven, followed by the RT30 with four. RS30 exhibited the largest proportion, constituting 42.9 % of the paths. Preferential flow paths with a width of  $10 < W < 20$  mm were exclusively found in RO30.

### 3.4. Influencing factors of soil infiltration capacity and dye-stained characteristic parameters

Correlation analysis showed that the soil infiltration capacity was positively correlated with both soil properties and root characteristics (Fig. 8). Specifically, IIR, SIR, and AIR showed a positive relationship with TP, MAP, and RLD ( $P < 0.05$ ), while SIR and AIR had a positive correlation with RSA ( $P < 0.05$ ). SIR exhibited a positive association with sand content ( $P < 0.05$ ). IIR, SIR, and AIR were significantly and negatively correlated with BD, IP, and clay content ( $P < 0.05$ ). Within the soil and root properties, there was a significant positive relationship between MAP and sand content, RLD, and RSA ( $P < 0.001$ ). In addition, the content of clay was associated positively with IP, and SOM content was positively correlated with RLD and RSA ( $P < 0.05$ ).





**Fig. 5.** Dye distribution in vertical soil sections of different vegetation types. Blue and red lines represent dye-stained coverage (DC, %) and relative deviation (RD, %), respectively.

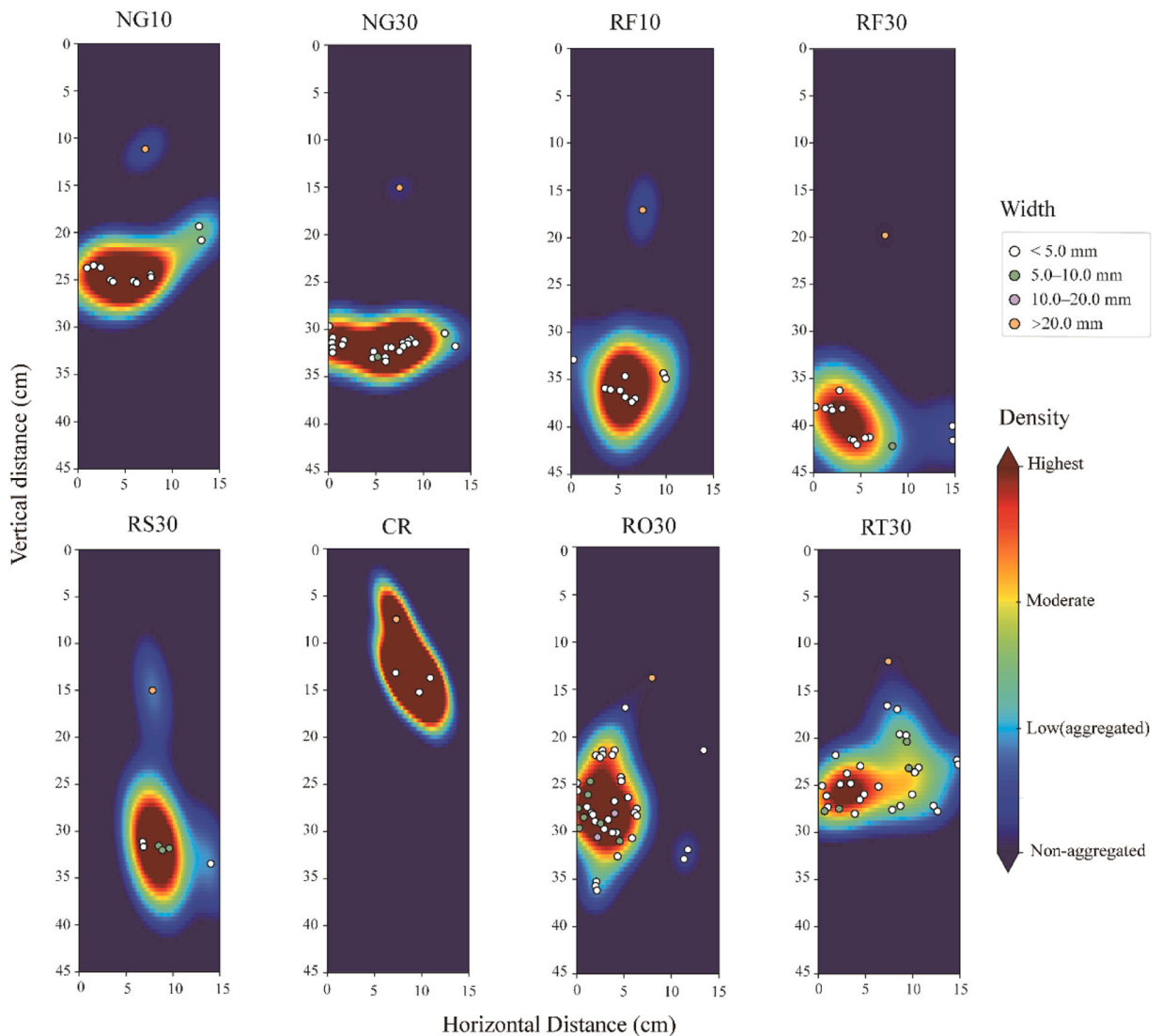
UniFr, TDC, and MSD was inversely correlated with BD and directly correlated with TP (Fig. 9). A good correlation ( $P < 0.05$ ) was examined among MSD, BD, and TP. Additionally, UniFr, TDC, and MSD showed significant positive correlation with SWC, MAP, and the content of sand ( $P < 0.001$ ), but was negatively associated with IP and the content of clay. LI was positively linked to the content of silt ( $P < 0.01$ ). The root characteristics also significantly affected the dye-stained characteristic parameters. There were positive

**Table 4**

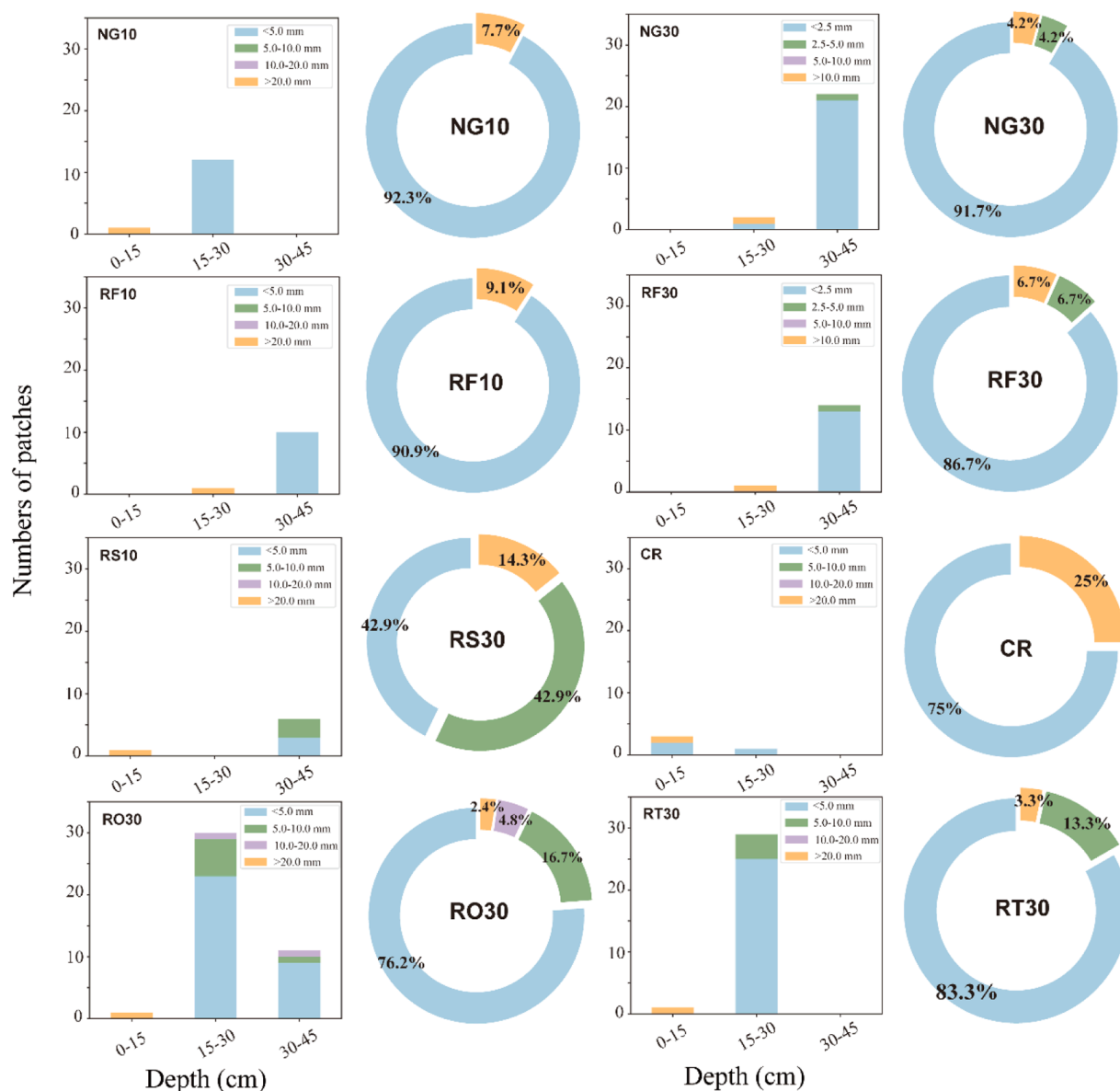
The parameters related to dye staining in the 0–45 cm soil layer among different vegetation types.

|                        | NG10   | NG30   | RF10   | RF30          | RS30   | RT30   | RO30          | CR            |
|------------------------|--------|--------|--------|---------------|--------|--------|---------------|---------------|
| MSD (mm)               | 24.90  | 31.57  | 35.80  | 42.08         | 33.58  | 25.43  | 34.32         | 16.83         |
| TDA (cm <sup>2</sup> ) | 329.73 | 450.59 | 513.04 | <u>595.11</u> | 447.69 | 352.17 | 405.62        | 221.46        |
| TDC (%)                | 48.87  | 66.78  | 76.04  | <u>88.20</u>  | 66.35  | 52.2   | 60.12         | 32.82         |
| UniFr (cm)             | 19.61  | 28.92  | 33.26  | <u>37.50</u>  | 27.09  | 19.54  | 21.06         | 12.88         |
| PFFR (%)               | 10.64  | 3.77   | 2.81   | 5.55          | 9.29   | 16.82  | <u>22.18</u>  | 12.85         |
| LI                     | 170.82 | 277.65 | 146.12 | 240.24        | 137.18 | 396.47 | <u>412.94</u> | 229.88        |
| CV (%)                 | 95.54  | 66.93  | 54.02  | 31.40         | 65.93  | 87.97  | 69.60         | <u>135.86</u> |

Note: Underlined entries indicate the highest values among different vegetation types. MSD: maximum stained depth; TDA: ratio of the dye-stained area; TDC: total dye-stained coverage; UniFr: uniform infiltration depth; PFFR: preferential flow fraction; LI: length index; CV: coefficient of variation



**Fig. 6.** Location distribution of stained patches with different widths in the vertical profiles of different vegetation types. The color gradient represents varying aggregation degrees of patches: deep blue indicates without aggregation, light blue indicates the occurrence of aggregation, yellow indicates moderate aggregation, and red–black indicates the highest degree of aggregation). Four different colors of points represent various widths of stained patches.



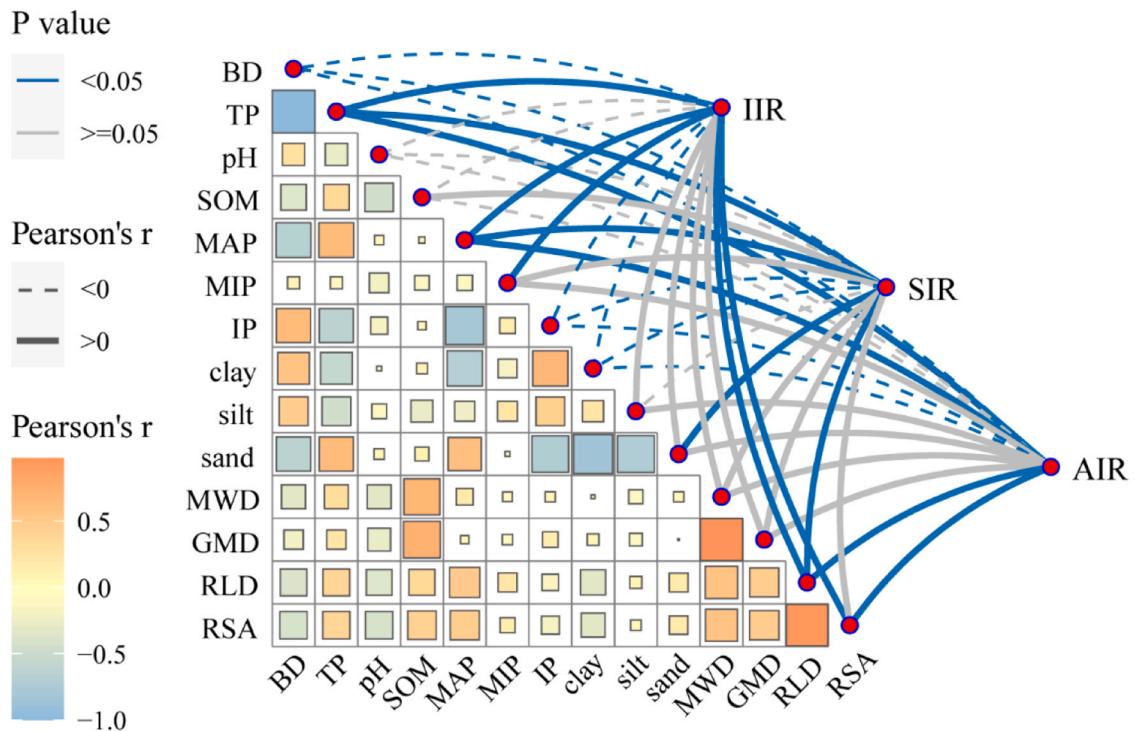
**Fig. 7.** Numbers and portion of stained patches with different widths in the vertical profiles of different vegetation types. Four different colors represent various widths of stained patches.

correlations between RLD and RSA with UniFr, TDC, PFFR, MSD, and LI. Specifically, RLD demonstrated the most significant association with UniFr, TDC, and, MSD, whereas RSA was substantially associated with MSD, LI, and PFFR.

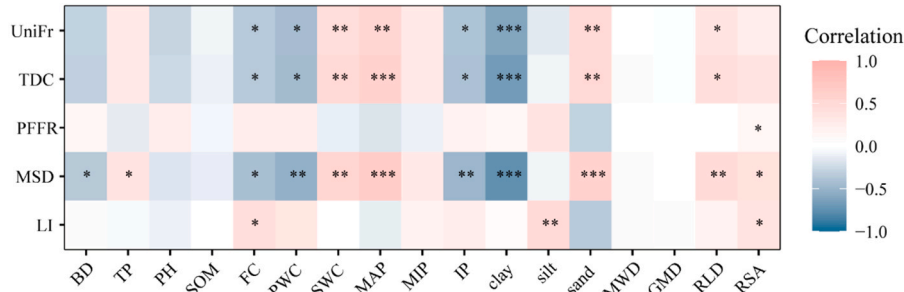
## 4. Discussion

### 4.1. Impact of vegetation restoration on soil infiltration capacity

The variability in soil infiltration following vegetation restoration is primarily attributed to changes in soil properties and the plant roots tratis. The evidence from this research suggest that the infiltration capacity of the topsoil is effectively enhanced after vegetation restoration (Zhu et al., 2020). Compared with cropland, the infiltration rate showed a significant increase in monocultured forestlands, mixed forestlands, shrublands, and grasslands. The infiltration capability of the RF30 was the highest, followed by that of the RS30. For different vegetation types, the highest infiltration capacity was in forestlands, and the lowest was in grasslands. However, the infiltration capacity of monocultured forestland was marginally greater than that of the mixed forestland. Furthermore, soil infiltration capacity improved as the number of years since vegetation restoration increased, which is in good agreement with previous results (Gu et al., 2019; Leung et al., 2018). Zhao et al.(2023) also found that the IIR and SIR increased with eucalyptus growing. This study



**Fig. 8.** Pairwise comparisons of soil infiltration capacity with soil properties and root characteristics. The color gradient denotes the Pearson's correlation coefficient. Edge type reflects the plus and minus of Pearson's statistic and the color of edge color indicates the statistical significance.



**Fig. 9.** Pearson correlation analysis on the dye-stained characteristic parameters including MSD, TDA, TDC, UniFr, PFFR, and LI, soil properties including BD, TP, SOM, MAP, MIP, IP, MWD, and GMD, and root characteristics including RLD and RSA.

demonstrated that vegetation restoration increased soil macroporosity and decreased ineffective porosity, except in the RT30. Moreover, a notable positive correlation was observed between soil porosity and infiltration capacity, particularly for macroporosity. Higher soil porosity enhances the infiltration capacity in semi-arid areas (Wu et al., 2016). Zhao et al. (2017) similarly observed that the aggregate microstructures in revegetated plots were better developed and more porous by quantifying the pore system.

Root characteristics were closely associated with soil macroporosity. Plant roots remarkably enhanced the number of soil macropores by forming root channels. Moreover, the decomposition process of plant root systems contributes to humus formation and promote the development of soil aggregates and macropores. NG30 exhibited the greatest aggregate stability and organic matter content at the 0–40 cm soil depth. Grasslands possess advantageous soil water conditions and shorter life cycles than forestlands and shrublands, which contribute to their greater potential for soil organic carbon sequestration (Burst et al., 2020). Roots also enmesh and rearrange soil particles while releasing exudates, causing physical, chemical, and biological changes in the soil that subsequently affect its infiltration capacity (Bronick and Lal, 2005). The soil clay content displayed a strong negative relationship with the soil infiltration capacity. Clay and silt particles showed a higher viscosity coefficient and soil water suction compared to sand, causing them to gradually swell during infiltration and impede water movement (Zhao et al., 2022). Conversely, sand content exhibited a strong positive relationship with macroporosity, which is explained by the fact that RT30 had the worst infiltration capacity among the vegetation restoration plots because it had the lowest sediment content and highest silt content. Soil texture and location are strongly

related. This could be attributed to fine soil particles eroding from the upper hillside and depositing on the forestland (Qiu et al., 2022).

#### 4.2. Effects of vegetation restoration on preferential flow

Overall, the infiltration patterns of the study area were primarily dominated by matrix flow. The percentage of dye-stained soil areas decreased with increasing soil depth, which aligns with the findings of previous studies (Kang et al., 2023). The vertical profiles of the dyed soil revealed that, water movement varied at different depths. The quantitative parameters (TDC, MSD, UniFr, PFFR, and LI) clearly reflect the traits of water flow paths in the vertical direction of soil after vegetation restoration. In contrast, the MSD, TDC, and UniFr values measured in the croplands were lower than in the plots after vegetation restoration, suggesting that vegetation restoration greatly promote the vertical infiltration. Among them, the greatest values were observed for RF30. The values of PFFR and LI point to a substantial development of preferential flow, thereby the degree of preferential flow in the vertical direction was highest in the mixed forestland, especially for RO30. Moreover, the quantity of preferential flow paths was also the greatest in mixed forestland. This result corroborates with that of Guan et al. (2024). With the increasing plantation age of grasslands and forestlands, both the degree and number of preferential flow paths rose, promoting preferential flow within the deeper soil layers.

The uneven distribution of soil conditions may cause the spatial variability of water flow paths. The clay content was negatively correlated with UniFr, TDC, and MSD, whereas the sand content showed the opposite trend. Higher clay contents lead to high soil bulk density, impeding the development of large pore structures and water infiltration. A regular pattern of depositional clay layers tends to form connected preferential flow paths (Maier et al., 2020), we found the degree of preferential flow was positively with clay and silt content. It also increased the uneven distribution of the dye in the soil, which was also found in Javaux et al. (2006). During the wetting and drying cycles, the swelling and shrinking of clay soils will further increase the uneven distribution of soil moisture (Qi et al., 2022).

This study showed that the UniFr, TDC, and MSD parameters were positively correlated with soil macroporosity, aligning with the findings of Zhang et al. (2022). Among all vegetation types, RF30 had the highest soil macroporosity and, root length density, and surface area at 0–40 cm soil depth, especially the presence of deeply rooted coarse roots. Research has shown that the presence of coarse roots led to a 30 % rise in macroporosity. Consequently, macropore flow was more pronounced in the vertical direction, leading to more effective water flow paths (Zhang et al., 2024) and greatly facilitate the vertical infiltration of dyes. The uniform infiltration depth was the deepest due to the dominance of roots and macropores as water flow paths and the uniform soil pore structure with an abundant distribution of large pores. This reflects the excellent permeability and vertical connectivity within the soil profile.

Root systems have a strong impact on the paths of water flow throughout the soil. It can be seen that UniFr, TDC, and MSD was closely associated with RLD ( $P < 0.05$ ). The complex root structure in the soil, which forms many large, well-connected pores (Li et al., 2023). Thus, the greater TDC is observed in the shallow soil layers. The distribution of dye displayed that the stained soil depth was relatively shallow in grasslands, since the fine root distribution and microbial activities of grasslands are concentrated near the soil surface. Fine root development in the soil has also become more uniform with the growth of grassland. In contrast, the distribution of preferential flow paths in forestlands was deeper than those in grasslands and croplands. Water moved vertically downward more efficiently through macropores in forest soil than in grassland soil (Alaoui et al., 2011). Moreover, root channel played an important role as preferential flow channels (Ni et al., 2019). RSA was highest in RO30 and was closely related to LI and PFFR; thus, root channels provided the pathways of preferential flow in RO30 mixed forestland. Plant roots strongly influenced the formation and continuity of macropores, which, in turn, affected the degree of preferential flow in soil profiles (Niu et al., 2015). Moreover, the degradation of roots can also form large holes that serve as preferential flow paths (Ghestem et al., 2011).

Notably, the bulk density and clay content between RT30 and NG10 showed no significant difference ( $P > 0.05$ ), whereas PFFR had higher values for RT30. From the perspective of soil structure in the dyed profiles, areas surrounded by rock fragments exhibited a larger proportion of active water movement in the dyed regions. This may be because *P. tabulaeformis* forests are commonly planted on slopes with high rock fragments coverage in a boreal region (Laine-Kaulio et al., 2015). The gaps between the rock fragments and soil provided preferential pathways for water or dye infiltration (Wang et al., 2024a, 2024b). Moreover, the dyed soil profile in the cropland showed that plowing could destroy most of the preferential paths within the root zone, which reduced the number of macropores linking the root zone with deeper soil, as well as changed in pore connectivity. However, the uneven distribution of dyes in croplands was much greater than that in the vegetation restoration plots. This may occur in soils with weak structure and obvious cracks, potentially causing water to flow laterally along cracks rather than infiltrating uniformly into the soil (Chen et al., 2024).

## 5. Conclusions

This study investigated the variations of soil infiltration capacity and preferential flow under different types and years of vegetation restoration and investigated their responses to soil and root properties in the hilly region of the Loess Plateau. The main findings are as follows:

- Vegetation restoration significantly increased soil infiltration capacity, with forestland having the highest, followed by shrubland and grassland. Monocultured forestland showed slightly greater infiltration capacity than mixed forestland. With an increase in vegetation restoration years, there was an observed increase in the infiltration capacity and quantity of preferential flow paths. The highest degree of preferential flow and the greatest number of preferential flow pathways were observed in mixed forestlands. The uniform pore structure and abundant large pores in RF30 facilitated the rapid and uniform water infiltration.



- All vegetation restorations increased MAP, and decreased IP, except for RT30. MAP, clay content, and RLD were key factors determining the soil infiltration capacity and dye-stained characteristic parameters. Root characteristics strongly impacted on the degree of preferential flow, while rock fragments and cracks were also playing significant roles.

The heterogeneity of soil structure and roots can influence water infiltration patterns, thereby affecting the effect of soil and water conservation. Therefore, it is important to consider suitable vegetation types, soil structure improvement, and effective water management strategies to ensure the sustainable implementation of vegetation restoration in the Loess Plateau.

For most rainfall events, the soil surface remains unsaturated and is not submerged, which may lead to different preferential flow patterns. Thus, future studies should focus on long-term monitoring and repeated measurements, using new methods like 3D soil moisture sensors to explore soil moisture dynamics under seasonal variations and different rainfall intensities, and analyze moisture movement patterns for more accurate assessment of the long-term impact of vegetation restoration on soil moisture dynamics.

#### CRediT authorship contribution statement

**Yang Mingyue:** Investigation, Data curation. **Liu Yuanhao:** Writing – review & editing, Visualization, Methodology, Investigation, Data curation. **Zhang Shuyu:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Zhao Guangju:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Data curation, Conceptualization. **Mu Xingmin:** Writing – review & editing, Writing – original draft. **Tian Peng:** Writing – review & editing, Visualization, Data curation.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Data availability

Data will be made available on request.

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